



ROBERT H. SMITH SCHOOL OF BUSINESS

BUDT704: DATA PROCESSING AND ANALYSIS IN PYTHON

A Data Analysis Report of Sleep, Health and Lifestyle.

TEAM: QHEMA (Section507-C)

TEAM MEMBERS: Qhamar Jahan LNU, Megha Mudigonda, Harini Gaddam, Emil George Mathew,
Aditya Deepak Kaushik

MENTOR: Prof. Peng Huang

INTRODUCTION

Sleep is one of the most crucial elements for maintaining optimal health, yet it is often sacrificed in the face of work, academic pressures, and other daily responsibilities. In our fast-paced world, the demands of professional and personal life can lead to poor sleep habits, leaving us feeling exhausted and vulnerable to stress. As college students, we too realise the consequences of sleep deprivation and as such, have decided to analyze the trends of sleep duration, quality of sleep, and sleep disorders.

We want to compare sleep quality and sleep duration with other qualities of a person's lifestyle and determine the impact of such factors by analyzing:

- What does the distribution of sleep duration look like?
- Is there a correlation between sleep duration and sleep quality?
- Do factors like the gender of a person, or stress level play a role in sleep disorders?
- Are lifestyle factors like BMI indicative of an effect on the duration of sleep?
- Can we predict the likelihood of sleep disorders in people based off the data we have?

BACKGROUND

Numerous studies find that quality and quantity of sleep have a severe impact on health. Chronic disease and lifestyle factors (e.g. work schedules) are associated with sleep deficiency, disruptions in circadian rhythm, and residual fatigue in an estimated 15%–30% of the general adult population [1]. While sleep deprivation or a lower duration and quality of sleep impact energy levels and health on the daily, a serious contribution to major public health concerns are sleep disorders.

Information from American Sleep Apnea Association suggests that the most recent data in the literature on the prevalence of sleep disorders found significant increases among some of the most common sleep disorders in the US from 2013-2016. According to the study, the prevalence of obstructive sleep apnea increased 41% over the study period, and narcolepsy increased 14% [2].

This project seeks to analyze data collected from individuals' daily routines, sleep patterns, and sleep quality. The dataset provides valuable insights into sleep-related metrics, as well as factors related to lifestyle, cardiovascular health, and sleep disorders. The aim of this report is to shed light on the impact of sleep disorders on poor sleep quality and insufficient sleep duration. In the context of a broader societal trend toward declining sleep quality and duration, this project focuses on identifying specific daily habits that influence individuals' sleep health as well as predicting the possibility of a sleep disorder.

Ultimately, the analysis in this report not only deepens the understanding of personal sleep health but also provides valuable guidance for individuals looking to make informed choices about their daily habits. By pinpointing the main factors that affect sleep quality and duration, individuals can take proactive steps to adopt behaviors that promote better sleep and support overall well-being and longevity.

OBJECTIVE

The objective of this project is to explore the relationship between sleep patterns, overall health, and lifestyle choices. By leveraging machine learning techniques, we aim to predict the likelihood of sleep disorders based on factors such as age, sleep duration, body mass index (BMI), occupation, stress levels, blood pressure, and existing sleep disorders. The goal is to provide an understanding of how these variables interact and influence sleep. Additionally, we also develop individuals at higher risk for sleep-related issues.

DATASET

The dataset we have decided to work with is the [Sleep, Health, and Lifestyle](#) dataset from Kaggle. The Sleep Health and Lifestyle Dataset consists of 374 records and 13 columns, offering a comprehensive view of individuals' sleep patterns and related lifestyle factors. Key features include sleep duration, quality, and the presence of sleep disorders such as Insomnia and Sleep Apnea, alongside lifestyle metrics like physical activity levels, stress, and BMI categories. It also captures important cardiovascular indicators, including blood pressure and heart rate, as well as daily steps. Each entry provides demographic details such as gender, age, and occupation, enabling in-depth analysis of how various factors impact sleep health and overall well-being. This dataset offers a rich resource for exploring the interplay between sleep, lifestyle habits, and health outcomes.

DATA PREPROCESSING & CLEANING

The data preprocessing phase involved several key steps to clean, transform, and structure the dataset for efficient analysis.

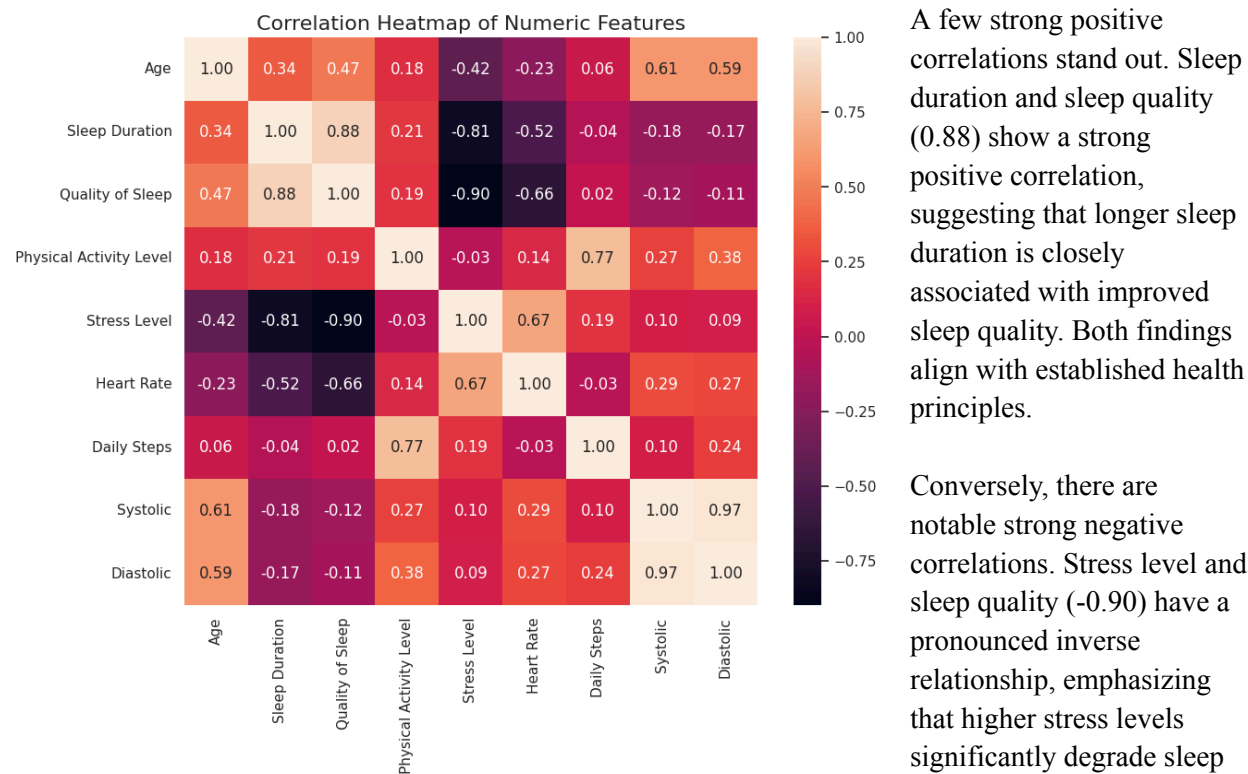
Handling Missing Values: One of the primary challenges in the dataset was the presence of missing values, particularly in the Sleep Disorder column. Missing values in this column were replaced with "No Sleep Disorder" to ensure that every individual had an assigned category. This was important for maintaining consistency and avoiding biases in downstream analysis and modeling.

Data Transformation: The Blood Pressure column initially contained systolic and diastolic values as a combined string (e.g., "120/80"). In order to process the information, this column was split into two separate columns: Systolic and Diastolic. The newly created columns were initially stored as object (string) types, but for numerical analysis, they were converted to integer data types to enable proper mathematical operations such as correlations and aggregations.

Categorical Data Handling: Some of the categorical variables required standardization for consistency. For example, the BMI Category values contained variations such as "Normal Weight" that were standardized to a single label ("Normal") for uniformity. This standardization process ensured that categories were represented uniformly across the dataset, reducing the risk of errors in analysis.

EXPLORATORY DATA ANALYSIS

To understand the dataset and its different features, we checked several different statistical functions. The first one being a correlation matrix. For this, we only consider the numerical feature data points.



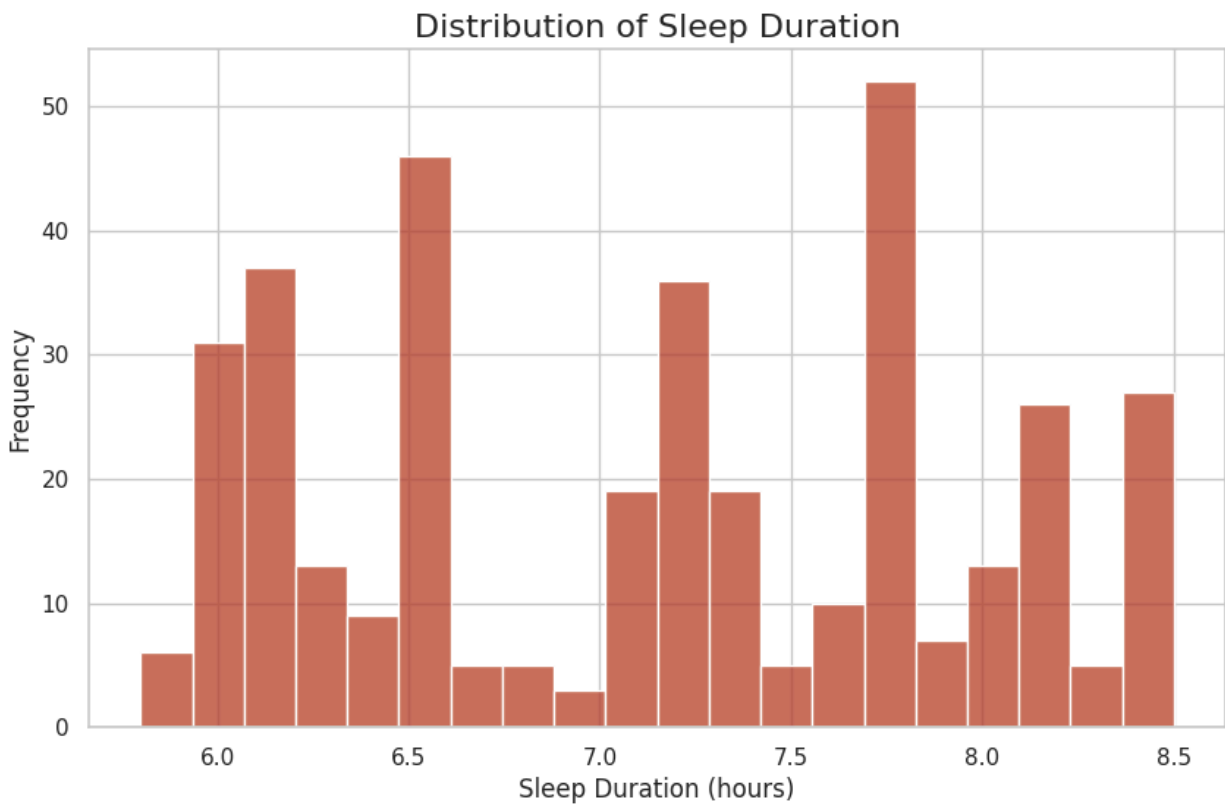
level and sleep duration (-0.81) highlight the negative impact of stress on the amount of sleep. Another striking observation is the negative correlation between heart rate and sleep quality (-0.66), indicating that poorer sleep quality is often associated with higher heart rates.

Moderate correlations are also evident. Age is moderately positively correlated with both systolic blood pressure (0.61) and diastolic blood pressure (0.59), showing a trend of increasing blood pressure with age. These findings align with well-documented age-related changes in cardiovascular health. Other correlations, such as physical activity level and daily steps (0.77), reflect expected associations where higher physical activity levels correspond with increased daily step counts.

DATA ANALYSIS AND INFERENCES

Question 1. What does the distribution of sleep duration look like for the population in this dataset?

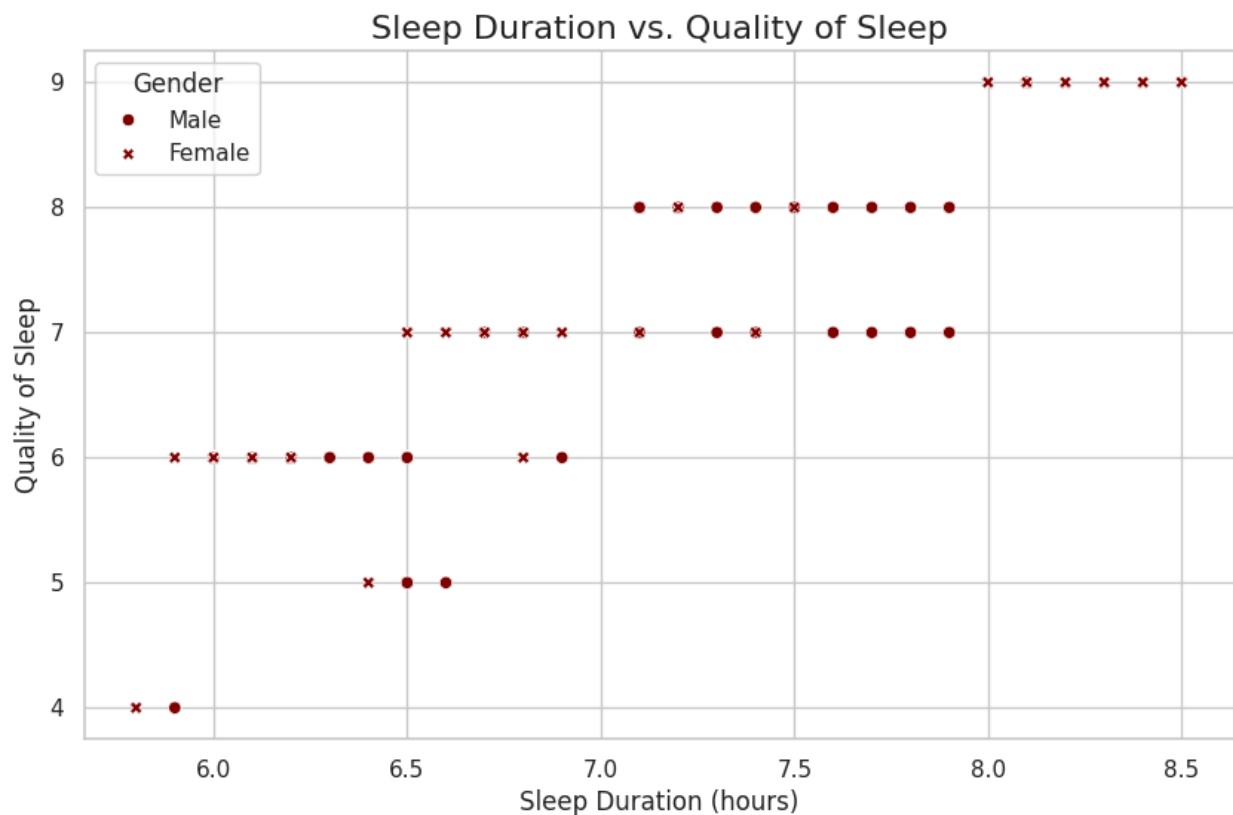
We check the histogram of sleep duration to understand the distribution. This histogram displays the distribution of sleep duration, revealing a bimodal pattern. This graph suggests that the most common sleep duration is around 6-8 hours.



Few individuals sleep less than 6 hours, representing those with insufficient sleep, while a small group sleeps more than 8 hours, reflecting extended sleep durations. The data highlights that while most individuals fall within or near the optimal range, noticeable subgroup experiences shorter sleep, warranting further investigation into underlying factors like lifestyle or stress.

Question 2. How does sleep duration impact sleep quality, and does this relationship vary by gender?

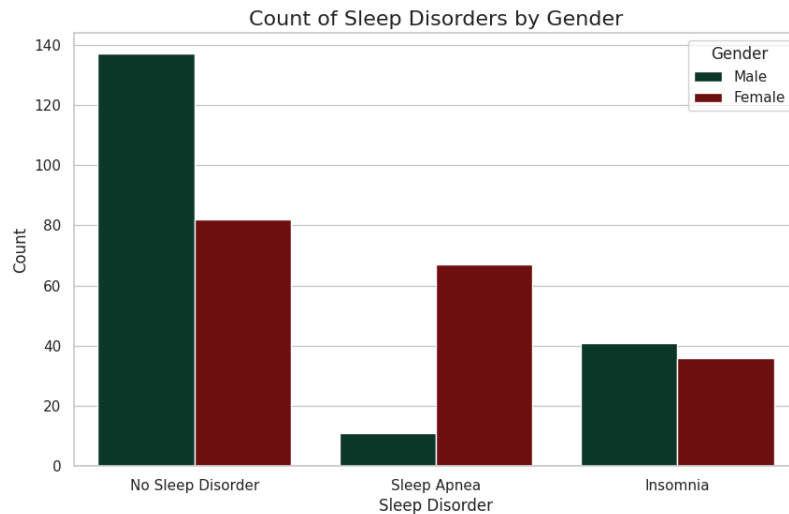
This scatter plot illustrates the relationship between sleep duration (x-axis) and sleep quality (y-axis), with data points differentiated by gender (male and female). Sleep quality is measured on a scale of 4 to 9. The overall trend suggests that longer sleep durations are generally associated with higher sleep quality, indicating a positive correlation between these two variables. Both males and females display similar patterns, with only slight variations in distribution.



Sleep quality improves consistently as sleep duration increases, with the highest quality reported around 8 to 8.5 hours of sleep. In contrast, individuals with shorter sleep durations (below 6.5 hours) are more likely to report lower sleep quality, typically in the range of 4 to 6. This trend emphasizes the importance of adequate sleep duration for achieving better sleep quality.

Interestingly, there is no notable clustering of data points by gender, indicating that the relationship between sleep duration and quality is consistent for both males and females. This insight suggests that interventions aimed at improving sleep duration would likely benefit individuals across genders in a similar way.

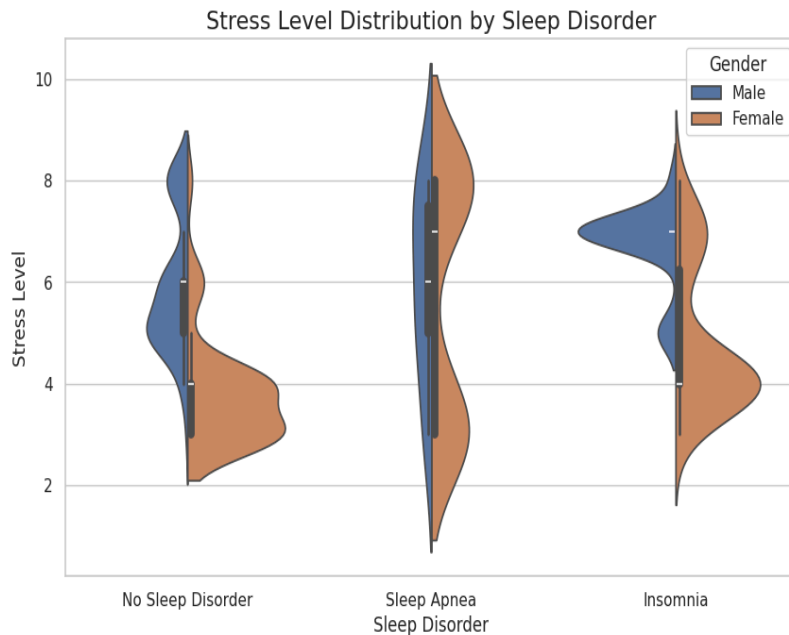
Question 3. What is the distribution of sleep disorders among males and females?



The bar graph shows the count of individuals with different sleep disorders categorized by gender. The majority of both males and females fall under the "No Sleep Disorder" category, with males having a significantly higher count than females. For "Sleep Apnea," females show a notably higher count compared to males, while the opposite trend is observed for "Insomnia," where males slightly outnumber females. This indicates that males are more likely to have

no sleep disorders but are more prone to insomnia, whereas females have a higher prevalence of sleep apnea. The data highlights the need for gender-specific approaches in understanding and addressing sleep disorders. This is in regards to the population for this dataset.

Question 4. How do stress levels vary across different sleep disorders for males and females?



The violin plot illustrates the distribution of stress levels across sleep disorders, with wider sections indicating higher densities. Individuals with "No Sleep Disorder" have lower, more concentrated stress levels, while "Sleep Apnea" and "Insomnia" show broader, more variable distributions. Females exhibit higher and more varied stress levels across all categories, suggesting gender-specific trends. This reinforces the link between elevated stress and sleep disorders, highlighting the need for targeted stress management strategies.

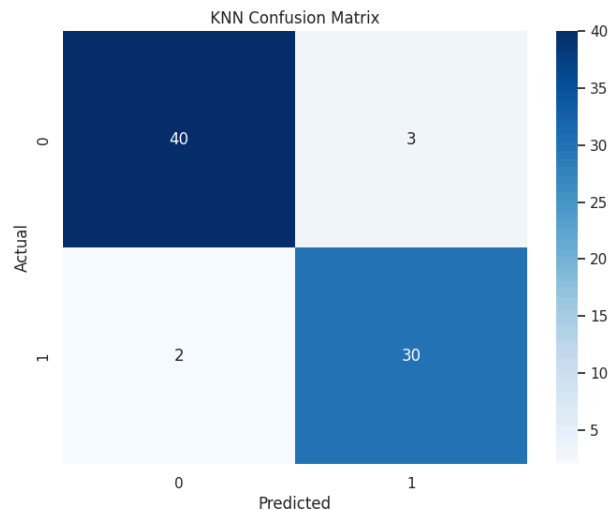
MACHINE LEARNING MODELS

KNN

KNN is a supervised learning computation that prognosticates a data point's similarity to other data points in the training dataset to predict its label. It works by examining the "k" nearest data points, or neighbors, according to their similarity, which is mostly determined by distance. It uses the majority label among these neighbors to predict the category for classification. Regression uses the average of the values of its neighbors to forecast a value. KNN can be slow when there are many data points to compare, but it is simple to understand and performs well with small datasets. To provide fair comparisons, the data must also be properly prepared, including by scaling features.

The `sleep_dataframe_final` dataset's target variable, `sleep_final`, is predicted using a K-Nearest Neighbors (KNN) classification model. Data preparation is the first step in the process, during which the target variable (y) is extracted independently and extra columns such as Person ID and the target column are eliminated from the characteristics (X). `StandardScaler` is used to standardize numerical characteristics so that they have a consistent scale in order to guarantee fair distance estimates in KNN. After that, the dataset is divided into training and testing sets, with a fixed random state for repeatability and 80% of the data going to training and 20% to testing. Using the scaled training data, KNN classifier (`KNeighborsClassifier`) is trained, and predictions are made on the test set.

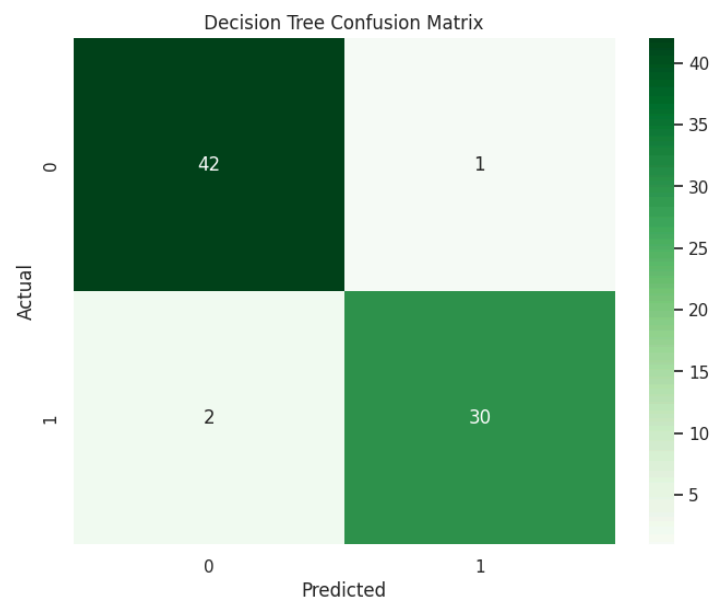
Metrics like accuracy, precision, recall, and F1 score are calculated to assess the model and give information on how effectively the classifier predicts across all classes. A categorization report also deconstructs these metrics for every class. To further examine model performance and identify misclassification areas, the confusion matrix, which shows the comparison between real and predicted classes, is presented as a heatmap. A framework for training, assessing, and visualizing the KNN classifier's performance on a structured dataset is provided by this procedure.



DECISION TREE

To predict the `sleep_final` target variable from the `sleep_dataframe_final` dataset, we use a Decision Tree Classifier. In order to guarantee consistent scaling, we first prepare the data by separating the features (X) from the target variable (y) and standardizing the numerical features using `StandardScaler`. We train a `DecisionTreeClassifier` with a fixed random state after dividing the dataset into training and testing sets, with 80% for training and 20% for testing. We use measures like accuracy, precision, recall, and F1 score, which are computed on the actual and predicted test labels, to assess the model's performance once it has been trained and predictions are made on the test set.

For every class, a classification report offers a thorough analysis of these parameters. Lastly, we plot a confusion matrix as a heatmap to illustrate the model's performance and examine the classifier's accuracy and mistakes across various classes.



RESULTS

In our model we get a higher accuracy for the Decision tree than KNN. Unlike KNN, decision trees can handle datasets with irrelevant or noisy features more efficiently since they automatically select the most informative features during the training phase. Furthermore, unlike KNN, which depends on standardized features for precise distance computations, Decision Trees are independent of feature scaling. As a result, the Decision Tree might be better equipped to adjust to the features of the dataset. Furthermore, whereas KNN may have trouble with overlapping classes or high-dimensional data, Decision Trees are better at identifying complex patterns in the data. The increased accuracy indicates that the Decision Tree is producing more accurate predictions in this situation and is more appropriate for the structure of our dataset.

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KNN Metrics:
Accuracy-----: 93.33%
Precision: 93.39%
Recall: 93.33%
F1 Score: 93.35%

Classification Report:

```

	precision	recall	f1-score	support
0	0.95	0.93	0.94	43
1	0.91	0.94	0.92	32
accuracy			0.93	75
macro avg	0.93	0.93	0.93	75
weighted avg	0.93	0.93	0.93	75

```

Decision Tree Metrics:
Accuracy-----: 96.00%
Precision: 96.02%
Recall: 96.00%
F1 Score: 95.99%

Classification Report:

```

	precision	recall	f1-score	support
0	0.95	0.98	0.97	43
1	0.97	0.94	0.95	32
accuracy			0.96	75
macro avg	0.96	0.96	0.96	75
weighted avg	0.96	0.96	0.96	75

LIMITATIONS

The analysis of sleep disorders in this project is subject to several limitations:

- The dataset is relatively small and has a lot of imbalances in terms of categories like "No Sleep Disorder", which are highly overrepresented. This tends to skew the results and impact the performance of predictive models.
- Furthermore, the data on occupation and stress levels could contain inherent biases in terms of a dataset containing a larger number of occupations as compared to others. This potentially reflects uneven sampling that may influence the findings.
- Another critical limitation is the absence of time series data, which prevents the analysis of trends or changes over time, limiting our ability to make longitudinal predictions.
- Despite these constraints, the machine learning models demonstrated moderate accuracy, indicating their potential for early detection of sleep disorders. Future studies with larger, more balanced datasets and additional temporal and demographic data are necessary to enhance the robustness and applicability of these models.

CONCLUSION

This project looks at how sleep patterns, lifestyle habits, and sleep disorders are all connected, and it came up with some pretty interesting findings. One of the coolest things was how effective machine learning, especially the Decision Tree model, turned out to be. It hit 96% accuracy, which is super impressive and shows how useful these tools could be for figuring out who's at risk of sleep disorders early on. It could really help with creating strategies to prevent sleep-related health problems.

One big thing we noticed was how much stress and sleep duration impact sleep disorders. Stress, in particular, had a huge negative effect on both the quality and length of sleep. This makes it clear that stress management—things like mindfulness, exercise, or even making workplaces less stressful—could go a long way in improving sleep and lowering the chances of sleep disorders.

We also found some gender differences that were pretty striking. Women in the dataset tended to have higher stress levels and were more likely to deal with sleep apnea, while men were more prone to insomnia. This kind of stuff shows that we need different solutions for different groups—what works for one won't necessarily work for the other.

WORKS CITED

- [1] James P Kiley, Michael J Twery, Gary H Gibbons, The National Center on Sleep Disorders Research—progress and promise, *Sleep*, Volume 42, Issue 6, June 2019, zsz105, <https://doi.org/10.1093/sleep/zsz105>
- [2] American Sleep Apnea Association—The State of Sleep Health in America 2023, <https://www.sleephealth.org/sleep-health/the-state-of-sleephealth-in-america/>
- [3] Sleep, Health, Lifestyle Dataset from Kaggle (<https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset/data>)